



Research article

Experimental and numerical investigation of process-induced recoil force in keyhole laser welding: Insights for validating multi-physics process simulations and modelling assumptions

Andreas Andersson Lassila ^{ID}*, Erik Lundell, Tobias Andersson ^{ID}, Dan Lönn ^{ID},
Kent Salomonsson ^{ID}, Rohollah Ghasemi ^{ID}

Virtual Manufacturing Processes, School of Engineering Sciences, University of Skövde, Kaplansgatan 11, Skövde, SE-541 34, Sweden

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ABSTRACT

Among the various driving forces involved in the molten pool during keyhole laser welding, the vaporization-induced recoil pressure is the dominant one. This study experimentally measured the process-induced recoil force during laser welding of aluminium and copper. A customized measurement setup was used to measure the specimen displacement caused by the recoil force, which was then determined by means of a finite element (FE) analysis. Furthermore, multi-physics computational fluid dynamics (CFD) models of the laser welding process were developed. After calibration, these models were used to predict the recoil force and its dependence on various process parameters. When only the recoil pressure acting on regions where vaporization occurs was considered, excluding the gaseous phases in the model, the total recoil force was underestimated. To account for that the formed gas contributes to the total recoil force as it rises and exits the keyhole, the total recoil force was calculated based on the predicted net mass flow due to vaporization and condensation. This simplified model showed good agreement between predicted and experimentally measured recoil forces, demonstrating that the observed consistent recoil force with increasing laser power may be due to a corresponding increase in the condensation rate. This highlights the importance of understanding the behaviour of the vaporized gas phase to determine appropriate simplifications and assumptions in laser welding process modelling. The findings of this study support the development and validation of multi-physics process models, further advancing knowledge of relevant modelling approximations.

1. Introduction

In recent years, laser welding has emerged as a cutting-edge technology for a wide range of joining processes. During laser welding, joints with minimal heat affected zones (HAZ) and large depth-to-width aspect ratios can be achieved due to the high intensity of the laser beam. This high intensity also reduces the thermal impact on the base material and surrounding components for laser welded joints. Furthermore, laser welding offers high process stability and efficiency, together with exceptional precision and promising conditions for automation [1]. Moreover, laser welding has enabled the joining of previously considered difficult-to-weld materials, such as copper (Cu) and aluminium (Al), as well as the joining of dissimilar materials [2]. Due to their excellent electrical conductivity, Al and Cu are commonly used as electrical

connectors and conductors in electric vehicle (EV) components, where laser welding is the preferred joining technology. For example, the laser welding of hairpin stators for electric motors [3,4], cell-to-busbar joints for battery pack assemblies [5,6], or busbars/current collectors for inverters or power electronics [7] are important processes that are widely considered in the laser welding literature and frequently used in the industry.

Although laser welding offers numerous advantages, the process of welding Cu and Al presents several challenges. For commonly used industrial near-infrared (NIR) lasers with wavelengths around 1 μm , both Cu and Al are highly reflective materials, exhibiting optical absorptivities near 5% at room temperature [8,9]. These materials often experience instability and defect formation, e.g. spatters [8,10], porosity [8,11], and fluctuations in penetration depth [8].

* Corresponding author.

E-mail addresses: andreas.andersson.lassila@his.se (A.A. Lassila), erik.lundell@his.se (E. Lundell), tobias.andersson@his.se (T. Andersson), dan.lonn@his.se (D. Lönn), kent.salomonsson@his.se (K. Salomonsson), rohollah.ghasemi@his.se (R. Ghasemi).

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Nomenclature

α	Accommodation coefficient
I	Incoming ray vector
n	Unit surface normal
R	Reflected ray vector
ΔH_v	Latenet heat of vaporization
ΔL	Extension of coarse mesh region
Δt	Processing time
δ_p	Displacement of measuring arm
$\dot{m}$	Net mass flow
γ	Specific heat capacity ratio
μ	Fluid viscosity
ν	Poisson's ratio
ρ	Density
ρ_{vap}	Vapour density
σ	Surface tension
σ_0	Surface tension coefficient
θ	Incident angle of laser beam
ϵ	Electrical conductance coefficient
$\vec{g}$	Acceleration due to gravity
$\vec{v}$	Fluid velocity field
A	Cross-sectional area of keyhole opening
a	Absorption rate
a_{condens}	Condensation factor
a_{plume}	Attenuation factor
c_p	Specific heat at constant pressure
c_v	Specific heat at constant volume
D	Average diameter of keyhole
d_p	Penetration depth
E	Young's modulus
F_{liq}	Fraction of liquid
$F_{r,\text{tot}}$	Total recoil force
h_f	Fluid enthalpy
h_{vap}	Heat transfer coefficient between vapour and keyhole wall
k	Thermal conductivity
k_{eq}	Equivalent stiffness of measuring arm
K_p	Isotropic permeability
L_f	Focal length
M	Molar mass
M_{vpr}	Magnification factor for upward vapour pressure
P	Laser output power
p	Fluid pressure
P_0	Ambient atmospheric pressure
P_L	Laplace pressure
P_r	Recoil pressure
$P_{S_{\text{max}}}$	Maximum pressure inside keyhole
P_{sat}	Saturation vapour pressure
P_{upward}	Upward vapour pressure
P_{vap}	Vapour partial pressure
R	Universal gas constant
r_f	Laser spot radius in focal plane
R_{II}	Principal radius of curvature

R_I	Principal radius of curvature
r_{lens}	Lens radius
r_{plume}	Radius of plume heat source
R_{spec}	Specific gas constant
S_M	Marangoni force
T	Surface temperature
t	Time
T_0	Reference temperature for surface tension
T_{amb}	Ambient temperature
t_{end}	Simulation time
T_{vap}	Vapour temperature
T_v	Vaporization temperature
v_f	Linear feed rate
v_{vap}	Vapour velocity
w_s	Weld seam width
x	Cartesian coordinate
y	Cartesian coordinate
z	Cartesian coordinate

understand the physical phenomena in the process zone. Furthermore, it is important to investigate the relationships between different process parameters and the resulting joint quality. If these investigations can be carried out with the help of numerical simulations, the time and cost of physical experiments can be significantly reduced. Furthermore, numerical simulations of laser welding processes can provide insights into physical quantities, such as velocity and temperature fields in the melt pool, as well as shape and dynamics of the keyhole, which are challenging to capture by in-situ observations. A commonly used simulation approach for laser welding processes is the use of multi-physics models, which through computational fluid dynamics (CFD) account for multiple physical phenomena, e.g. laser beam propagation, hydro-dynamic fluid and vapour flow, and the effects of base material vaporization. Here, the finite volume method (FVM) or the finite difference method (FDM), is often used in combination with the volume of fluid (VOF) method for free surface tracking, to predict the velocity, pressure, and temperature fields [12]. This multi-physics approach, using CFD models to simulate the laser welding process, has been extensively discussed in the literature. For example, in the study conducted by Otto et al. [13], a multi-physics CFD model was employed to investigate the laser welding of Cu and Al, taking into account both the liquid and gaseous phases. Using this model, the authors revealed critical physical mechanisms for porosity formation, and investigated the effects of different process parameters on keyhole formation and joint geometry. In the research conducted by Zenz et al. [14], the authors utilized a multi-physics CFD model to study the keyhole laser welding of Cu, and investigated keyhole formation, spatter, and porosity formation. In this model, an approach called the Mass-of-Fluid method was implemented to track the motion of both the liquid and gaseous phases, instead of the commonly used VOF method. Similar to the models described in [13,14], Florian et al. [15] utilized a multi-physics CFD model that considered both the liquid and gaseous phases of the base material to comprehensively investigate porosity formation during laser welding of Cu. As Florian et al. [15] emphasized, the modelling of both liquid and gaseous phases implies that the vaporization-induced recoil pressure arises as a result of the physical formulation of the vaporization process. However, this could lead to increased computational complexity, so a common simplification is to model only the liquid phase and introduce the effects of vaporization (recoil pressure, heat transport between gas and liquid, and vapour condensation) as boundary conditions on the melt pool and keyhole wall. Hayat et al. [16] utilized this simplification, and studied the effects of laser beam shaping by comparing the influence of circular

1.1. Numerical simulation of laser welding processes

In order to increase process stability and reduce defect formation during laser welding of Cu and Al alloys, it is crucial to better

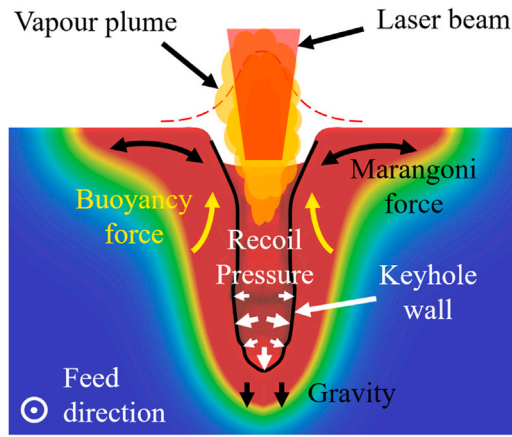


Fig. 1. Schematic representation of the main driving forces present in the melt pool during keyhole laser welding.

and tailing beam shapes on melt pool dynamics and porosity formation during laser welding of aluminium AA1060 battery busbars. A multi-physics CFD model was employed to simulate the welding process, demonstrating that keyhole necking and collapsing had a significant impact on the porosity formation. In the investigation performed by Wang et al. [17], the authors employed a multi-physics CFD model to study the mitigation of spatters during superimposed core and ring-shaped laser beam welding of Al partial penetration bead-on-plate joints. In this model the recoil pressure was introduced as a boundary condition on the melt pool, neglecting the gaseous phases in the model formulation. This allowed a time-efficient investigation of how different core/ring power ratios affected the proportion of spatter, as well as the joint geometry and the size of the keyhole opening.

As already mentioned, when employing multi-physics CFD models to simulate the laser welding process, it is crucial to take into account the various driving forces acting in the melt pool. Fig. 1 shows a schematic representation of the main driving forces, including the Marangoni force, buoyancy force, recoil pressure, gravity, and the effects from the vapour plume. The Marangoni force refers to the phenomena that the fluid flows from regions of low surface tension to regions of high surface tension, driven by the surface tension gradient [18]. All of these driving forces influence the fluid flow in the melt pool, and for keyhole laser welding the recoil pressure is the most dominant force [19,20]. Due to the high intensity in the laser spot during laser welding, the surface temperatures can rise well above the vaporization temperature of the base material. This high surface temperature leads to melting and vaporization of the base material. The recoil pressure is the counter force, exerted on the melt pool, by the escaping atoms due to this vaporization [21]. This recoil pressure leads to a depression in the melt pool and the formation of a keyhole. Due to this large influence of the recoil pressure on melt pool dynamics and fluid flow, it is crucial for an accurate modelling and simulation of the laser welding process, to know both the magnitude of this pressure and its variation as a function of the surface temperature.

A frequently used method for modelling the recoil pressure in multi-physics CFD simulations of laser welding processes is to assume that the recoil pressure, P_r , is proportional to the saturation vapour pressure, P_{sat} , according to

$$P_r = \alpha P_{\text{sat}} \quad (1)$$

where the proportionality constant, α , corresponds to the accommodation coefficient [20]. According to Semak and Matsunawa [20], the value of the accommodation coefficient depends on the ambient pressure and should be assigned a value between 0.55 and 1, which corresponds to vaporization in vacuum and at a high ambient pressure,

respectively. Furthermore, Semak and Matsunawa [20] state that the accommodation coefficient for many practical laser applications is close to the minimum value of 0.55, and for laser welding applications a value of this magnitude is commonly used, cf. [22,23]. The saturation vapour pressure is commonly expressed as a function of the surface temperature, T , by the Clausius-Clapeyron equation [21], according to

$$P_{\text{sat}} = P_0 \exp \left(\frac{\Delta H_v}{(\gamma - 1) c_v T_v} \left(1 - \frac{T_v}{T} \right) \right) \quad (2)$$

where P_0 is the ambient atmospheric pressure, T_v is the vaporization temperature of the base material, c_v is the specific heat at constant volume, ΔH_v is the latent heat of vaporization, and $\gamma = \frac{c_p}{c_v}$ is the specific heat capacity ratio for the vapour. From Eqs. (1) and (2) the recoil pressure can be expressed as

$$P_r = A \exp \left(B \left(1 - \frac{T_v}{T} \right) \right) \quad (3)$$

where the coefficients A and B are given by Eq. (4), according to

$$\left. \begin{aligned} A &= \alpha P_0 \\ B &= \frac{\Delta H_v}{(\gamma - 1) c_v T_v} \end{aligned} \right\} \quad (4)$$

These coefficients are commonly used as parameters for calibrating the model on the basis of experimental results, cf. [11,16]. Typically, this calibration is performed indirectly by adjusting the modelled recoil pressure to match experimental melt pool or keyhole shapes. For a more direct and precise calibration, experimental measurements are needed, with some studies reported in the literature. For example, in the experimental investigation performed by Kuznetsov [24], the recoil momentum during laser irradiation of metallic targets was measured by the utilization of a ballistic pendulum and a piezo-electric force sensor. In this investigation, an empirical relation, demonstrating the dependence of recoil momentum on ambient pressure, was presented. Knorovsky and MacCallum [25] performed both direct force measurements using a piezo-electric force sensor, and indirect pressure measurements using microphones to record the sound, or acoustic emissions, from the process during pulsed laser spot welding of 304 stainless steel. For the direct piezo-electric force measurements, the authors reported a recoil force magnitude of approximately 8 mN. For the indirect measurements, the authors reported recoil forces between 2.2–11 mN for pulse energies between 2–7 J, respectively. In the research conducted by Mizutani et al. [26], the recoil pressure distribution was estimated based on keyhole curvature and depth, measured from in-process X-ray observations during laser irradiation of a liquid zinc (Zn) bath. Furthermore, the estimates were verified by direct measurements of recoil pressure with a precision electric balance, during laser spot welding of solid Zn specimens. Mizutani et al. [26] reported experimentally measured recoil pressures of 4–5 kPa, corresponding to recoil forces of 1.1–1.4 mN for a keyhole diameter of 0.6 mm. Cullom et al. [27], investigated the resulting recoil force in a laser powder bed fusion (L-PBF) process for 304L stainless steel specimens. The exerted recoil force was quantified by means of measurements in the frequency domain, using modal analysis. In this investigation, the authors found that the recoil forces were below 0.5 mN when welding in heat conduction mode, and above 10 mN when welding in keyhole mode.

1.2. Motivation and overview

A common simplification when modelling keyhole laser welding processes is to only include the effects of vaporization in terms of a temperature dependent recoil pressure locally on the melt pool, but the gaseous phases from this vaporization process are excluded [15]. However, the formed gas (which is excluded from the simulation) produce an additional pressure, i.e. the vapour partial pressure, acting

on the keyhole wall. In order to compensate for this excluded pressure, the recoil pressure needs to be increased at the local point. Consequently, the recoil pressure inside the keyhole may be overestimated and thus the keyhole dynamics will be different. Therefore, it is important to compare experimental measurements of recoil forces with predicted values from simulations to determine which simplifications are acceptable in the modelling process. As previously discussed, there are some experimental investigations reported on the measurement of recoil force during laser welding processes. However, less attention has been paid to the comparison between experimentally measured and modelled recoil forces in multi-physics CFD simulations. To our knowledge, this is primarily due to a lack of experimental data. Therefore, the relationships between the recoil force and various laser process parameters, need to be further investigated using both physical experiments and numerical process simulations.

In the present study, the process-induced recoil force during laser welding of Cu and Al specimens was experimentally measured. The choice of studying Cu and Al is based on that these materials exhibit similar processing behaviours, such as low optical absorption and high thermal conductivity, while differing in properties like density and heat capacity. These characteristics make them appropriate for demonstrating the applicability and generalizability of the proposed approach and method. A hybrid experimental-numerical approach was utilized, in which the displacement of the considered specimens caused by the exerted recoil force was indirectly measured. The recoil force was calculated from the displacement data using a finite element (FE) model. Furthermore, multi-physics CFD models were developed for the laser welding process for both Cu and Al specimens. These CFD models were calibrated using experimentally measured joint geometries obtained from weld seam cross-sections. After calibration, the CFD models were used to predict the vaporization-induced recoil pressure and its dependence on various process parameters. In addition, a total process-induced recoil force was predicted for various process parameters and compared with the experimentally measured values. Worth to notice is that this process-induced recoil force is in this work considered as the total reaction force acting on the specimen caused by the net mass flow escaping the keyhole due to vaporization.

Therefore, the contribution of this study is to present a hybrid approach to measure the recoil force in the melt pool during laser welding processes, as well as an investigation of how the recoil force is affected by various process parameters and sample materials. Furthermore, this work provides some insights into how the process-induced recoil force can be estimated from multi-physics CFD simulations, given certain assumptions or simplifications. Moreover, the present study highlights the importance of understanding the physical phenomena in the keyhole to determine appropriate simplifications and assumptions in laser welding process modelling, and also to give some insights whether the gaseous phases need to be included in order obtain accurate values of the local pressure field. The findings support the development and validation of multi-physics process models, further advancing knowledge of relevant modelling approximations. Although this study focuses on laser welding of Al and Cu, the findings can be generalized to other materials, and the methods used are applicable to various laser welding processes as well as laser-based additive manufacturing.

The remaining parts of this article are structured as follows. The experimental method for measuring the recoil force, and the modelling approach for the considered multi-physics CFD models are described in Section 2. The results and findings obtained from this study are presented in Section 3, followed by a discussion of the results in Section 4. Finally, the conclusions from this research, together with suggestions for future work are presented in Section 5.

2. Hybrid approach and model description

In order to measure the recoil force during laser welding experiments of Al and Cu specimens, a hybrid experimental-numerical approach was employed. A customized measurement setup was used

Table 1

Laser source and optics configuration.

TruDisk 6001 with PFO 33-2	
Adjustable power [W]	120–6000
Wavelength [nm]	1030
Fibre core diameter [μm]	100
Fibre ring diameter [μm]	400
Focal length [mm]	255
Collimation length [mm]	150
Aspect ratio [–]	1:1.7
Focal spot diameter (core) [μm]	170
Focal spot diameter (ring) [μm]	680
Beam parameter product [mm mrad]	4
Field size PFO [mm^2]	180 × 112

to measure the specimen displacement caused by the exerted recoil force. The displacement was then used to calculate the resulting recoil force, by using a FE model of the measurement setup. Furthermore, multi-physics CFD models of the laser welding process were developed for both Al and Cu specimens, which were employed to investigate the relationships between the recoil force and various laser process parameters.

2.1. Laser welding experiments

The laser welding experiments were performed with a 5-axis laser machine (TruLaser Cell 3000 from Trumpf, Ditzingen, Germany), equipped with a focusing optics (PFO 33–2 from Trumpf, Ditzingen, Germany) which was used for laser beam delivery and for the realization of different welding paths through remote laser welding. The utilized laser source was a 6 kW solid-state disk laser (TruDisk 6001 from Trumpf, Ditzingen, Germany). The generated laser light was delivered to the focusing optics through a 2-in-1 fibre, consisting of a core and a ring fibre with diameters of 100 μm and 400 μm , respectively. With this 2-in-1 fibre, static beam shaping (BrightLine Weld from Trumpf, Ditzingen, Germany) was also considered, where different beam shapes were generated by superimposing the core and ring beams, each with a different peak power. Further details on the optical configuration and the laser source are shown in Table 1.

Fig. 2a shows a schematic representation of the experimental setup, where the specimen is mounted on a customized measuring arm, consisting of a stainless steel plate and a Bakelite indicator. As shown in Fig. 2a, the steel plate is fixed to the machine frame as a self-supporting cantilever beam, and the indicator is adhesively bonded to the steel plate using epoxy. The cross-sectional dimensions of the steel plate and the indicator are 1.0 mm × 25.6 mm and 2.15 mm × 10.0 mm, respectively. This setup allowed for the measurement of the deflection of the steel plate due to the recoil force acting on the specimen during the laser welding experiments. The indicator acts as an extension of the steel plate from the point of force application and thus provides an enlarged deflection. During the welding process, the recoil force, F_r , will deflect the steel plate and the point P on the indicator will be displaced to a point P' with the corresponding displacement δ_p . This displacement was measured using a laser triangulation displacement sensor (optoNCDT 1420 from Micro-Epsilon, Ortenburg, Germany) and the recoil force was determined by

$$F_r = k_{eq} \delta_p \quad (5)$$

where k_{eq} represents the equivalent stiffness of the measuring arm, determined from the FE analysis described in Section 2.2. The cross-sectional dimensions of the steel plate, along with the free length of the cantilevered plate and the length of the indicator, result in a maximum displacement lower than one twentieth of the Rayleigh length for the considered laser configuration.

The considered specimens were Al and Cu plates, each with a thickness of 4 mm, and cut to dimensions of 15 mm × 25 mm, as shown

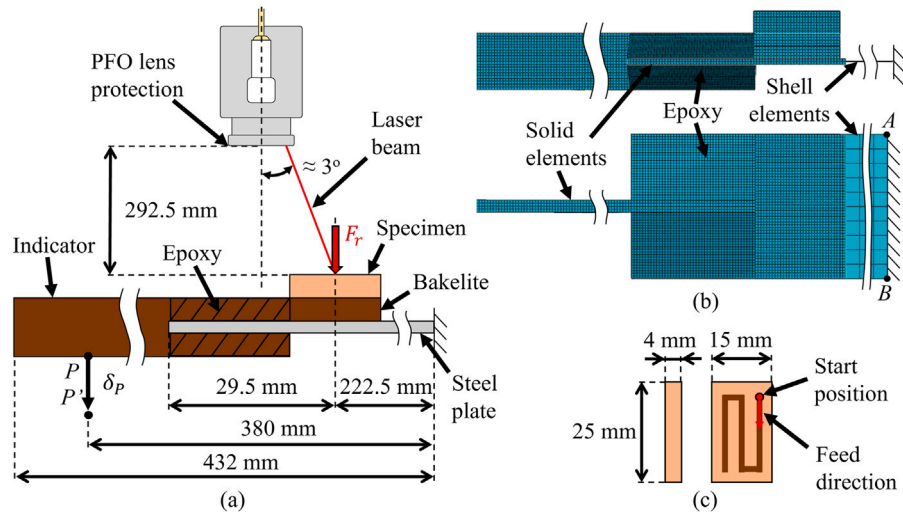


Fig. 2. Schematic representation of: (a) experimental setup, (b) finite element (FE) model for the calculation of recoil force, and (c) the considered specimen and welding path.

Table 2

Chemical composition for EN AW 1050A and EN CW004A (wt.%) [28,29].

	Al	Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti
EN AW 1050A	≥99.50	≤0.25	≤0.40	≤0.05	≤0.05	≤0.05	–	≤0.07	≤0.05
	Cu	Bi	Pb	O					
EN CW004A	≥99.90	≤0.0005	≤0.005	≤0.040					

in Fig. 2c. The Al plates were made of the material type EN AW 1050A and the Cu plates were made of the material type EN CW004A (Cu-ETP). Table 2 lists the chemical composition for EN AW 1050A and EN CW004A. As shown in Fig. 2c, the considered welding path was an S-shaped curve, consisting of five connected line segments, with a total weld seam length of 53 mm. This welding path was used to increase the laser processing time, and thereby extend the data acquisition time for measuring the deflection of the measuring arm. At a feed rate of $v_f = 200$ mm/s, this weld seam length corresponds to a laser processing time of $\Delta t = 0.265$ s. To obtain a smooth approaching in the laser-material interaction, the laser power was ramped up from 0%–100% over the first 0.5 mm, and ramped down from 100%–0% over the last 0.5 mm. To prevent back reflections of laser light into the optics, the laser welding process was performed with an offset of 25 mm between the focusing optics and the weld seam position. Due to the f-theta lens configuration for the focusing optics, this results in a laser beam inclination angle of approximately 3°, as shown in Fig. 2a. To prevent heat transfer between specimens and the measuring arm, all contact surfaces were thermally insulated with Bakelite. Adhesive tape was used to bond the specimens to the Bakelite, as well as to bond the Bakelite to the measuring arm. This prevents the measuring arm from being deformed by the heat from the laser welding process. Prior to the experiments, all specimens were cleaned using Petroleum benzine and Acetone.

In the experiments, five different parameter setups were considered for each material, whereby the laser output power, P , and the linear feed rate, v_f were varied for two different beam shapes. Fig. 3 illustrates the considered beam shapes, which are characterized by the proportion of peak power delivered to the core and ring fibre. The analysis considers peak power proportions for the core of 100% and 65%. In all experiments, the focal position was set on the top surface of the specimen, and welding was performed without shielding gas or pressurized air to suppress the vapour plume. Table 3 shows the considered parameter setups, labelled A1–A5 for Al specimens, and B1–B5 for Cu specimens. Each parameter setup in Table 3 was repeated three times to ensure reproducibility and account for variability. The recoil force was only measured for parameter setups A1–A4 and B1–B4.

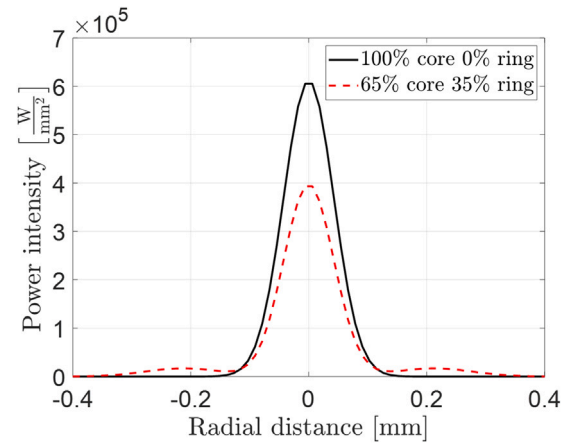


Fig. 3. Illustration of the considered beam shapes consisting of 100% core 0% ring intensity, and 65% core 35% ring intensity.

Table 3

Parameter setups considered in laser welding experiments.

Parameter setup	Material [–]	P [W]	v_f [mm/s]	Core proportion [%]
A1	Al	3000	200	100%
A2	Al	4000	200	100%
A3	Al	5000	200	100%
A4	Al	5000	200	65%
A5	Al	5000	400	100%
B1	Cu	4000	200	100%
B2	Cu	5000	200	100%
B3	Cu	6000	200	100%
B4	Cu	6000	200	65%
B5	Cu	6000	400	100%

After the laser welding experiments, micrographs of the welded specimens were prepared to investigate the joint geometry. The specimens were cut across the weld seam as shown in Fig. 4, and the weld

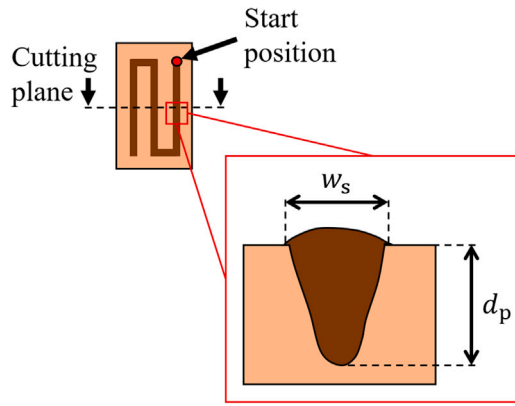


Fig. 4. Schematic illustration of the considered cross-sections and joint geometry.

seam cross-section of the first vertical line in the S-shaped welding path was used for further analysis. This cross-section was used to ensure an accurate comparison with the simulated results concerning the increased temperature of the specimen during the process, and that the welding process was simulated for a shorter welding path. After cutting, the specimens were mounted in resin, ground, polished and etched, then the joint geometry was measured using an optical microscope (Stemi 2000-C from Zeiss, Oberkochen, Germany). Here, the penetration depth, d_p , and the weld seam width, w_s , were measured according to Fig. 4.

2.2. Determination of process-induced recoil force

The process-induced recoil force was determined by Eq. (5), where k_{eq} is the equivalent stiffness of the measuring arm. This equivalent stiffness was determined from a dynamic implicit FE analysis of the measuring arm using Abaqus/Standard 2021. Fig. 2b shows the discretized FE model of the measuring arm consisting of indicator, steel plate, epoxy, specimen, and Bakelite insulation. The indicator, specimen, Bakelite insulation, and the first 38 mm of the steel plate (see Fig. 2b) were meshed using linear hexahedral solid elements with reduced integration of type C3D8R and an element size of 0.5 mm. The epoxy was meshed using linear wedge elements of the type C3D6 and an element size of 0.5 mm, and the last part of the steel plate (from specimen to fixed position) was meshed using quadratic quadrilateral shell elements of the type S8R and an element size of about 2.5 mm. For the nodes on the line A-B in Fig. 2b, both the translational and rotational degrees of freedom were set to zero. The recoil force, F_r , was applied as a constant pressure acting on an area of 2.0 mm $\times$ 25 mm, centred on the top surface of the specimen. Thus, no fluctuations in the recoil force were considered. This pressure was applied for a time period equal to the laser processing time, Δt . The time increment used in the simulation was 0.01 s, which corresponds to about 7% of the period time for the natural oscillations of the measuring arm. The FE model was calibrated by comparing experimentally measured displacements for known loads, with the corresponding displacements predicted by the model. Moreover, the experimentally measured eigenfrequencies of the measuring arm were compared with the corresponding predicted frequencies with a good fit. Here, Young's modulus for the steel plate was used as a model parameter, and was calibrated to $E = 160$ GPa. A possible reason for the determined low value is most likely due to flexibilities in the clamping. The material parameters, corresponding to density, ρ , Young's modulus, E , and Poisson's ratio, ν , used in the FE model are presented in Table 4. Here, the values for density are based on the presented references but have been corrected using measurements of the mass and volume of the specific specimens.

Table 4

Material parameters used in the FE-analysis for determination of recoil force.

Material	ρ [$\frac{kg}{m^3}$]	E [GPa]	ν [–]	Ref.
EN AW 1050A	2700	69	0.33	[28,30]
EN CW004A	8834	117	0.33	[29,31,32]
Stainless steel	7724	160	0.29	[33]
Bakelite	1355	7	0.35	[34]
Epoxy	1150	3.1	0.3	[35]

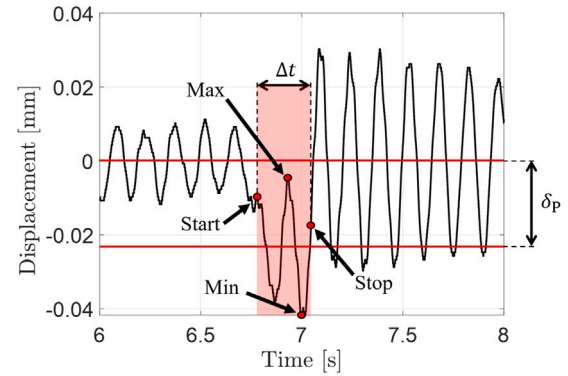


Fig. 5. Illustration of how the displacement, δ_p , was determined from the measured displacement signals from the laser welding experiments. This measurement signal corresponds to one of the repeated experiments for parameter setup B4.

Fig. 5 shows a typical measurement signal (corresponding to the measurement signal for one of the repeated experiments for parameter setup B4) for the displacement measurement during the laser welding experiments. Initially, before the laser welding process, the measuring arm oscillates at its natural frequency of about 6.9 Hz. This is due to very small environmental disturbances acting on the sensitive measuring arm. At the beginning of the laser welding process, the resulting recoil force on the specimen pushes the measuring arm downwards, causing it to oscillate around a new equilibrium position. FE-simulations showed that steady-state was reached after the first oscillation even though a small pre-oscillation was introduced representing the small environmental disturbances observed from experiments. Therefore, steady-state was assumed after one oscillation also in the experiments. As seen in Fig. 5, the laser processing time, Δt , corresponds to two oscillations of the loaded measuring arm. This criterion was used to determine the weld seam length, and consequently, the processing time for the laser welding experiments. After the laser welding process, the oscillations stabilize around the unloaded equilibrium position with an increased amplitude. According to Fig. 5, δ_p was defined as the displacement of the equilibrium position, which was calculated as the average of the minimum and maximum displacement during the time period Δt . Worth to notice is that the purpose of the experimental setup was not to measure the high dynamics of the process, e.g. a rapidly changing gas flow or keyhole fluctuations. The sole purpose was to measure the average recoil force acting on the specimen during the process time due to the laser-material interaction. This means that the frequencies for keyhole fluctuations, or other high frequency dynamics, cannot be measured. The oscillations shown in Fig. 5 only correspond to natural oscillations for the forced vibrations of the measuring arm due to the recoil force from the laser welding process.

After determination of δ_p for the parameter setups in Table 3, the FE model was used to determine the corresponding recoil force, by adjusting the magnitude of the applied pressure in the model until a predicted displacement of magnitude δ_p was obtained. This was done for one of the total three repetitions of each parameter setup. The obtained forces and corresponding displacements were then used to

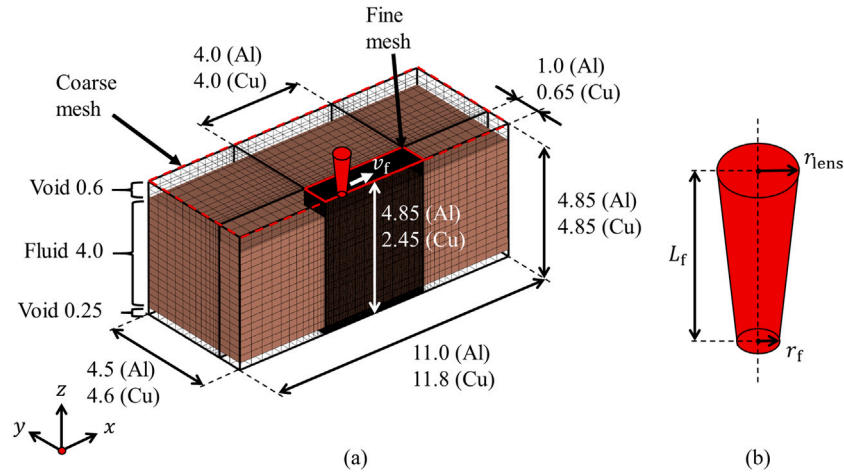


Fig. 6. Schematic illustration of: (a) the computational domain for the multi-physics computational fluid dynamics (CFD) model (dimensions in mm), and (b) the modelled laser beam.

calculate the equivalent stiffness, k_{eq} , by linear regression. Eq. (5) was then finally used to calculate the recoil force for the remaining repeated experiments.

2.3. Model description

For the prediction of process-induced recoil force and to investigate the relationships between recoil force and various laser process parameters, a multi-physics CFD model was developed in the commercial CFD software FLOW-3D 12.0 Release 7 Update 3 and the corresponding weld module FLOW WELD 3.0.1.1.8 Release 7. The model is based on the following assumptions:

- (1) The liquid metal in the melt pool is modelled as an incompressible Newtonian fluid.
- (2) The gaseous phases of the base material are not directly considered, but their effects are accounted for in the model through imposed boundary conditions, i.e. recoil pressure, mass loss due to vaporization, and thermal effects from the vapour plume.
- (3) The power intensities for both core and ring beam are assumed to follow a Gaussian distribution.

In the model, only the first 3 mm of the first vertical line of the weld path in Fig. 2c was simulated. This corresponds to approximately 2 mm of steady-state keyhole welding and ensures a time-efficient prediction. Furthermore, the process can be simplified as symmetric because the temperature field and pressure field spread out in a symmetrical manner from the centre of the heat source. Therefore, symmetry of this linear welding path was assumed, thereby allowing only half of the weld seam to be modelled. During the initial modelling work, models with and without symmetry were compared without any relevant differences in either penetration depth, joint width or process behaviour. The computational domain for the CFD model, consisting of two nested mesh regions composed of Eulerian mesh cells with different cell sizes, is illustrated in Fig. 6a. In order to obtain a sufficient resolution for the laser-material interaction, the laser process zone was discretized in a fine mesh region with a cell size of 0.025 mm. The size of this fine mesh region is 4.0 mm × 1.0 mm × 4.85 mm for the Al model, and 4.0 mm × 0.65 mm × 2.45 mm for the Cu model. These sizes were selected based on the size of the laser processing zone and the region where melt and material flow occur. As shown in Fig. 6a, the fine mesh region is also nested with a coarse mesh region, with a cell size of 0.25 mm. This coarse mesh region was only modelled to act as a heat sink, accounting for the thermal diffusion from the process zone to the surrounding material in the specimen. Since only the thermal diffusion was considered here, a larger cell size was used, and thus a

reduced number of elements, in order to reduce the calculation time. The extension of the coarse mesh region, ΔL , with respect to the fine region, in the x and y direction was calculated according to

$$\Delta L = 2 \sqrt{2 \frac{k}{\rho c_p} t_{\text{end}}} \quad (6)$$

where k , ρ , and c_p , are the thermal conductivity, mass density, and the specific heat at constant pressure for the specimen material [36]. This distance represents the thermal diffusion length, which is the extent to which heat diffuses through the specimen material during the simulation time, t_{end} . Therefore, the size of the coarse mesh region was determined by this length in order to avoid any boundary effects on the heat flow. The determined size of this coarse mesh region was 11.0 mm × 4.5 mm × 4.85 mm for the Al model, and 11.8 mm × 4.6 mm × 4.85 mm for the Cu model. The maximum time step for the CFD simulations was 3.75×10^{-7} s.

The boundary conditions assigned to the fine mesh region are symmetry conditions for all boundaries, i.e. boundaries in x , y , and z directions. For the coarse mesh region a wall condition with a specified ambient temperature of $T_{\text{amb}} = 293$ K is used for the boundaries in x direction, and for the boundary with maximum y coordinates. For the boundary with minimum y coordinates (i.e. $y = 0$) a symmetry condition is used. For the boundaries in the z direction, a pressure condition with a specified pressure of 0.1 MPa (atmospheric pressure) and $T_{\text{amb}} = 293$ K, is used. This is due to the modelled void region atop and underneath the specimen, according to Fig. 6a.

As illustrated in Fig. 6b, the laser beam was modelled with a conical shape with a spot radius in the focal plane of $r_f = 0.34$ mm, a lens radius of $r_{\text{lens}} = 19.6$ mm, and a corresponding focal length of $L_f = 255$ mm. The spot radius, r_f , corresponds to the focal $1/e^2$ spot radius for the ring fibre. The intensity for the two considered beam shapes were modelled according to Fig. 3. The laser beam moves with a constant feed rate, v_f , in the positive x direction. The simulations showed a steady-state behaviour in terms of penetration depth and weld seam width after 1–1.5 mm, therefore, a 3 mm simulated weld seam length was used. Furthermore, the model also incorporates the multiple reflections within the keyhole by the utilization of a ray tracing method. In this approach, the laser beam is discretized into multiple rays. When an incoming ray vector, denoted as I , hits the keyhole wall a small fraction of the energy is absorbed, while a larger fraction is reflected as a reflected ray vector, R . Provided that these vectors are normalized, the reflected ray vector is given by

$$R = I - 2(I \cdot n)n \quad (7)$$

where n represents the unit surface normal for the keyhole wall [11]. Additional energy is absorbed when the reflected ray hits the keyhole

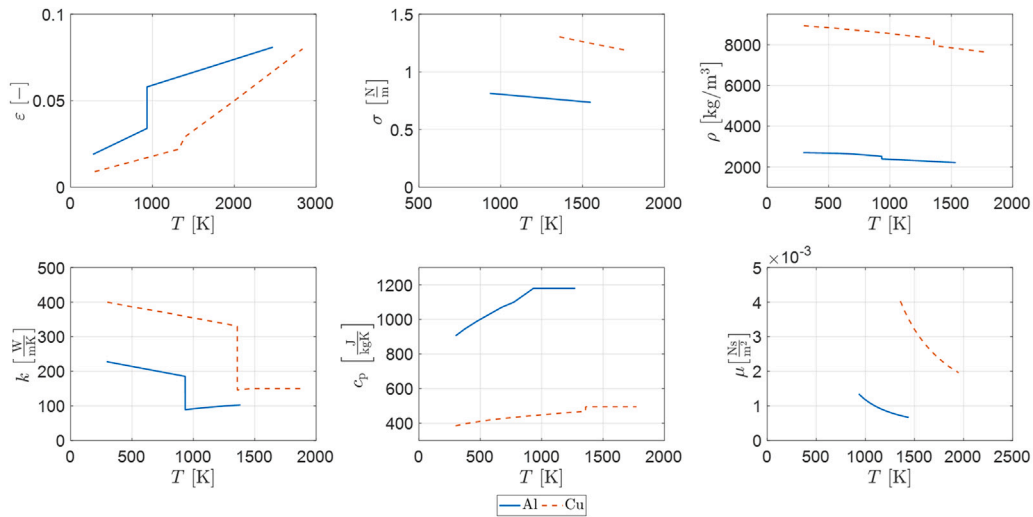


Fig. 7. Thermo-physical properties for aluminium (Al) (blue solid line) and copper (Cu) (orange dashed line), compiled from [37–39] for the Al model, and [15,36,37,40,41] for the Cu model.

wall at a different location. In the model, this absorption-reflection process is repeated until the energy density of the ray is lower than a predetermined attenuation factor, or if a predetermined maximum number of reflections is reached. The absorption rate associated with the laser-material interaction was modelled with Fresnel absorption, where the absorption rate, $a(\theta)$, is a function of the incident angle, θ , for the laser beam and is given by

$$a(\theta) = 1 - \frac{1}{2} \left(\frac{1 + (1 - \epsilon(T) \cos \theta)^2}{1 + (1 + \epsilon(T) \cos \theta)^2} \right) - \frac{1}{2} \left(\frac{\epsilon(T)^2 - 2\epsilon(T) \cos \theta + 2 \cos^2 \theta}{\epsilon(T)^2 + 2\epsilon(T) \cos \theta + 2 \cos^2 \theta} \right) + \epsilon_0 \quad (8)$$

where $\epsilon(T)$ is the temperature dependent electrical conductance coefficient for the base material and ϵ_0 is a constant used to adjust the absorption rate against experimental measurements [17,37]. The temperature dependency for $\epsilon(T)$ is shown in Fig. 7, for both the Al and Cu model, and $\epsilon_0 = 0$ was used for both models.

To predict the dynamics and morphology of the melt pool, as well as the resulting pressure and temperature fields, the conservation equations for mass, momentum, and energy, according to Eq. (9) to Eq. (11), were utilized [42].

$$\frac{\partial}{\partial t} (\rho) + \vec{\nabla} \cdot (\rho \vec{v}) = 0 \quad (9)$$

$$\frac{\partial}{\partial t} (\rho \vec{v}) + \vec{\nabla} \cdot (\rho \vec{v} \otimes \vec{v}) = \vec{\nabla} \cdot (\mu \vec{\nabla} \otimes \vec{v}) - \vec{\nabla} p - \frac{\mu}{K_p} \vec{v} + \rho \vec{g} \quad (10)$$

$$\frac{\partial}{\partial t} (\rho h_f) + \vec{\nabla} \cdot (\rho \vec{v} h_f) = \vec{\nabla} \cdot (k \vec{\nabla} T) \quad (11)$$

In Eq. (9), ρ corresponds to the fluid density and $\vec{v}$ denotes the fluid velocity field. Moreover, in Eq. (10), μ and p represent the fluid viscosity and the resulting fluid pressure, respectively, K_p is the isotropic permeability given by the Kozeny-Carman equation, and $\vec{g}$ denotes the acceleration due to gravity [42]. In Eq. (11), h_f is the fluid enthalpy, T is the temperature of the fluid, and k is the thermal conductivity. In Eq. (9) to Eq. (11), $\vec{\nabla} = [\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}]$ represents the Nabla operator and $\otimes$ denotes the outer product.

The VOF method was employed for free surface tracking, including the interface between liquid and void, as well as the resulting surface deformations. Here, the location and morphology of the interface boundary are determined by calculating the fraction of liquid, F_{liq} , for each time increment and mesh cell in the computational domain. Mesh cells with $F_{liq} = 0$ only contains void, and cells with $F_{liq} = 1$ only contains liquid. Mesh cells with values of $0 < F_{liq} < 1$ contains both

Table 5

Material properties used in phase transition models for Al and Cu.

Material property	Al	Cu	Ref.
Solidus temperature [K]	933.15	1346.5	[15,37,38]
Liquidus temperature [K]	933.55	1347	[15,37,38]
Latent heat of fusion $\left[\frac{\text{kJ}}{\text{kg}} \right]$	386	208.7	[43,44]
Vaporization temperature [K]	2792	2835	[37,43]
Latent heat of vaporization $\left[\frac{\text{MJ}}{\text{kg}} \right]$	10.8	4.73	[43,44]
Vapour specific heat $\left[\frac{\text{J}}{\text{kg K}} \right]$	896	384.6	[37,43]

liquid and void, and thus the liquid-void interface. The location of the liquid-void interface, and consequently, the location and morphology of the melt pool surface and the keyhole wall, can then be determined from [42]

$$\frac{\partial F_{liq}}{\partial t} + \vec{\nabla} \cdot (F_{liq} \vec{v}) = 0 \quad (12)$$

The gaseous phases were modelled as a void and are not directly considered in the model, but their effects are accounted for by imposing boundary conditions on the melt pool. The mass loss due to vaporization was modelled as a mass transfer between liquid and void, where the mass flow per unit area is given by

$$\dot{m}_a = \frac{\alpha}{\sqrt{2\pi R_{\text{spec}} T}} (P_{\text{sat}} - P_{\text{vap}}) \quad (13)$$

where $R_{\text{spec}} = (\gamma - 1)c_v$ denotes the specific gas constant for the metal vapour, α is the accommodation coefficient, and P_{sat} and P_{vap} are the saturation pressure and the void vapour partial pressure, respectively [40]. The saturation pressure was determined according to Eq. (2). The vaporization-induced recoil pressure, P_r , acting on the melt pool and keyhole wall, was modelled according to Eq. (1), where α and γ were calibrated to fit experimental results. In this process, the values for α and γ were calibrated to achieve a good fit between predicted and experimentally measured joint geometries. The accommodation coefficients and the specific heat capacity ratios for the Al and Cu material were calibrated to $\alpha_{\text{Al}} = 0.62$ and $\alpha_{\text{Cu}} = 0.43$, and $\gamma_{\text{Al}} = 1.67$ and $\gamma_{\text{Cu}} = 1.53$, respectively. Table 5 lists the material properties used to model the solid-liquid and liquid-gas phase transitions.

In addition to the vaporization-induced mass loss and recoil pressure, the intense vaporization during the laser welding process causes vapour to flow out of the keyhole, generating a vapour plume. From this vapour plume, heat is transferred to the top of the melt pool and to the keyhole wall. Moreover, the vapour plume may attenuate the laser

Table 6
Model parameters for the vapour plume model.

Parameter	Al	Cu
r_{plume} [mm]	0.47	0.35
a_{plume} [-]	0.24	0.78
a_{condens} [-]	0.99	0.99
h_{vap} [$\frac{\text{W}}{\text{m}^2 \cdot \text{K}}$]	2.25×10^6	8.73×10^6
T_{vap} [K]	6400	8000
M_{vpr} [-]	0	0.75

beam through absorption, i.e. inverse Bremsstrahlung [1]. To consider these effects, a Gaussian distributed heat source was used to model the vapour plume, which acts on a circular area with radius r_{plume} at the top surface of the melt pool [37]. The attenuation factor, a_{plume} , which represents the proportion of the total laser power absorbed by the vapour plume, determines the power for this heat source. To account for heating due to condensation of vapour on the keyhole wall, a condensation factor, a_{condens} , was used to specify how much of the vaporization energy from the vapour that should be returned to the keyhole wall [37]. Furthermore, heat transfer between the hot vapour in the keyhole and the colder keyhole wall was modelled with a heat transfer coefficient, h_{vap} , and a modelled vapour temperature, T_{vap} . Following [1], the heat transfer coefficient, h_{vap} , was estimated as

$$h_{\text{vap}} = \frac{48}{11} \left(\frac{k}{D} \right) \quad (14)$$

where D is the average diameter of the keyhole. Also, an upward vapour pressure, due to the ascending vapour from the bottom of the keyhole, was considered in the model. This upward vapour pressure, P_{upward} , is applied to the keyhole wall as a dynamic pressure according to

$$P_{\text{upward}} = P_{\text{S}_{\text{max}}} M_{\text{vpr}} \cos \phi \quad (15)$$

where $P_{\text{S}_{\text{max}}}$ is the maximum pressure inside the keyhole, M_{vpr} is a magnification factor to adjust the pressure value, and ϕ is the angle between the upward vapour direction and the free surface normal [37, 45]. The parameters r_{plume} , a_{plume} , a_{condens} , h_{vap} , T_{vap} , and M_{vpr} were used to calibrate the model against experimental results, and the values used in this work are listed in Table 6.

Laplace pressure and Marangoni forces were considered in the model by the use of a temperature dependent surface tension, $\sigma(T)$, as shown in Fig. 7. The Laplace pressure, P_L , was formulated using

$$P_L = \sigma_0 \left(\frac{1}{R_I} + \frac{1}{R_{II}} \right) \quad (16)$$

where R_I and R_{II} are the principal radii of curvature of the curved liquid surface [40]. Furthermore, the Marangoni force, S_M , was determined as

$$S_M = \sigma_0 - \frac{d\sigma}{dT} (T - T_0) \quad (17)$$

where σ_0 is the surface tension coefficient, i.e. the surface tension of the liquid metal at the melting point, and T_0 is the reference temperature [45]. T_0 was set to the melting temperature for the corresponding materials. For the model description above, the thermo-physical properties for Al and Cu are assumed as temperature dependent, as shown in Fig. 7. These thermo-physical properties are compiled from [37–39] for the Al model, and [15,36,37,40,41] for the Cu model. The temperature dependent absorption given in [41], for the Cu material, is slightly adjusted to improve agreement with experimental results.

3. Results

In Section 3.1 the results for the experimental measurements of recoil forces for both Al and Cu specimens are presented. Moreover, Section 3.2 presents the calibration of the CFD models for the Al and Cu

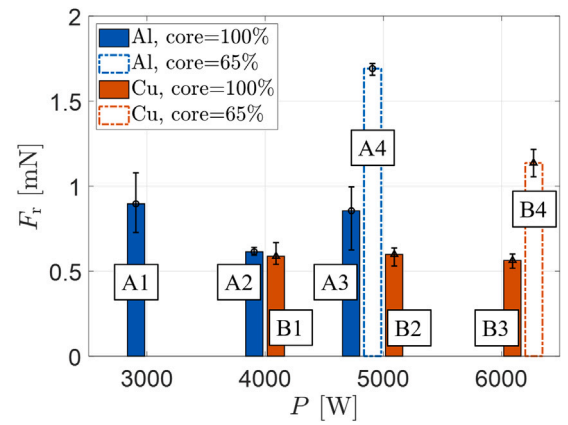


Fig. 8. Experimentally measured recoil forces for parameter setups A1-A4 and B1-B4. The figure shows the mean values from three repeated experiments for each parameter setup with the corresponding maximum and minimum values illustrated as error bars.

specimens, and Section 3.3 gives a comparison between predicted and experimentally measured recoil forces. Furthermore, the relationships between recoil forces and various process parameters are examined in this section.

3.1. Experimentally measured recoil forces

The experimentally measured recoil forces for parameter setups A1-A4 and B1-B4, as detailed in Section 2.1 and Table 3, are shown in Fig. 8. Here, the mean values from three repeated experiments for each parameter setup are shown, with the corresponding maximum and minimum values illustrated as error bars. For the Al specimens, the mean value for the recoil force shows consistent values for parameter setups A1 and A3, with measured mean values of 0.90 mN and 0.86 mN, respectively. For parameter setup A2, a decreased mean recoil force was observed, with a measured mean value of 0.61 mN. The measurements for the Cu specimens show a similar behaviour, with consistent mean recoil forces for parameter setups B1-B3, but with generally lower forces compared to the Al specimens. The measured mean recoil forces for parameter setups B1-B3 are 0.59 mN, 0.60 mN, and 0.56 mN, respectively.

When the peak power proportion of the core is decreased from 100% to 65%, corresponding to parameter setup A4 for the Al specimens and parameter setup B4 for the Cu specimens, the mean recoil force is significantly increased, both for Al and Cu specimens. As shown in Fig. 8, the measured mean recoil forces for parameter setups A4 and B4 are 1.69 mN and 1.14 mN, respectively.

3.2. Computational fluid dynamics (CFD) model calibration

The multi-physics CFD models described in Section 2.3, were used to simulate the laser welding process using the parameter setups presented in Table 3. For model calibration, the predicted penetration depths (d_p) and weld seam widths (w_s) were compared with the corresponding experimentally obtained values. Fig. 9a–b show a comparison between the predicted joint geometry, and the corresponding joint geometry extracted from experimentally obtained micrographs for parameter setups A2 and A4. Presented here are cross-sectional views normal to the feed rate direction, and the predicted joint geometry is visually represented by the variable melt region. In the case of parameter setup A2, the CFD model underpredicts the joint depth by less than 1%, while overpredicting the weld seam width by 3%. For parameter setup A4, the joint depth and weld seam width are underpredicted by 4% and 10%, respectively. Fig. 10a–b shows a similar comparison between predicted and experimentally obtained joint geometries for parameter setups B3

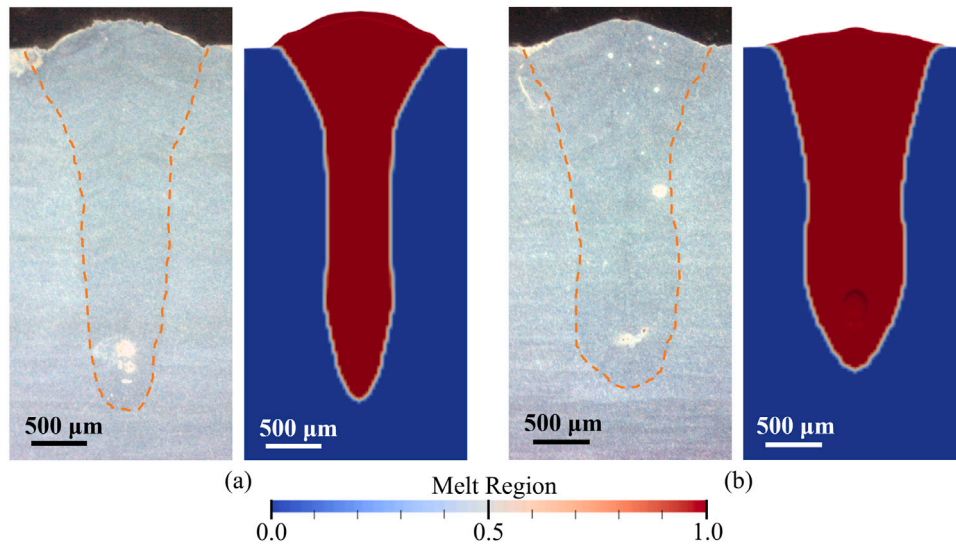


Fig. 9. Comparison between predicted joint geometry, and the corresponding joint geometry extracted from experimentally obtained micrographs for: (a) parameter setup A2 ($P = 4000$ W, $v_f = 200$ mm/s, core = 100%), and (b) parameter setup A4 ($P = 5000$ W, $v_f = 200$ mm/s, core = 65%).

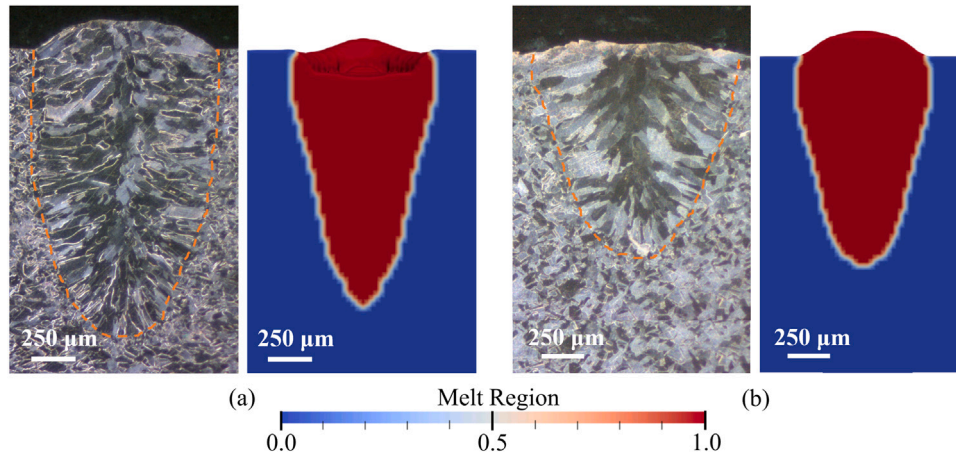


Fig. 10. Comparison between predicted joint geometry, and the corresponding joint geometry extracted from experimentally obtained micrographs for: (a) parameter setup B3 ($P = 6000$ W, $v_f = 200$ mm/s, core = 100%), and (b) parameter setup B4 ($P = 6000$ W, $v_f = 200$ mm/s, core = 65%).

and B4. Considering parameter setup B3, the CFD model underpredicts the joint depth by 16% and the weld seam width by 21%. For parameter setup B4, the joint depth is overpredicted by 8%, and the weld seam width is underpredicted by 26%.

Fig. 11 shows a summary of the calibrated results for all parameter setups in Table 3. Here, the mean values of d_p and w_s for three repeated experiments for each parameter setup are shown, and the corresponding minimum and maximum values are illustrated as error bars. Furthermore, the corresponding predicted values are presented for comparison. The predicted values for d_p and w_s were obtained from representative cross-sections in the steady-state region of the simulations where some variability is expected but were not considered to influence the obtained values to any larger extent. This comparison shows that the Al model reproduces the experimentally obtained values with relative errors below 4% for d_p and below 15% for w_s . As shown in Fig. 11, the relative errors for the Cu model is larger compared with the Al model. For the Cu model the relative error is below 25% for d_p and below 26% for w_s . Based on the results presented in Figs. 9a–b and 10a–b, as well as the comparison presented in Fig. 11, the models were considered as representative for predicting the joint geometry for the considered range of process parameters.

3.3. Numerical investigation of process-induced recoil forces

After calibration, the CFD models were used to predict and investigate the process-induced recoil forces during the laser welding process. Fig. 12 presents the predicted temperature fields and the regions where a recoil pressure is applied to the keyhole wall for six different time steps during the laser welding process for parameter setup A3. Here, the upper figures show the predicted temperature fields as contour plots, while the lower figures highlight in red the regions where the applied recoil pressure is larger than zero. As seen in Fig. 12, a recoil pressure is only applied to surfaces with surface temperatures at or above the vaporization temperature.

The recoil force in the z -direction was calculated as an average for 20 randomly selected time steps, by multiplying the recoil pressure component in the z -direction with the corresponding cell areas with an applied pressure. These resulting average recoil forces for parameter setups A1–A4 and B1–B4 range from 0.02 mN to 0.2 mN. This is significantly lower than the experimentally measured forces for the same parameter setups, ranging from 0.52 mN to 1.72 mN. The large discrepancy may be explained by the absence of gaseous phases of the base material after vaporization in the CFD model. In the model, the recoil pressure is considered as the counter force acting on the melt pool

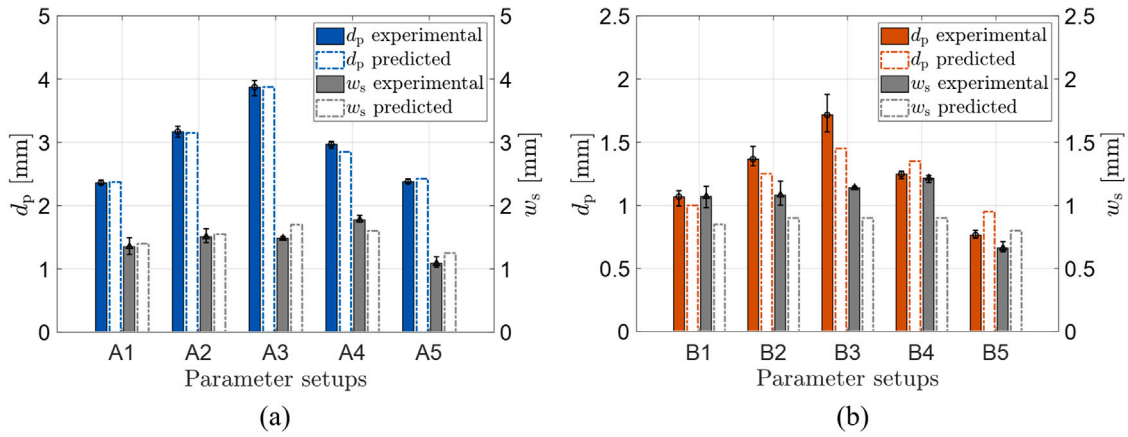


Fig. 11. Comparison between predicted joint geometry, and the corresponding joint geometry extracted from experimentally obtained micrographs for parameter setups (a) A1-A5, and (b) B1-B5. The figure shows the mean values of d_p and w_s for three repeated experiments for each parameter setup with the corresponding minimum and maximum values illustrated as error bars.

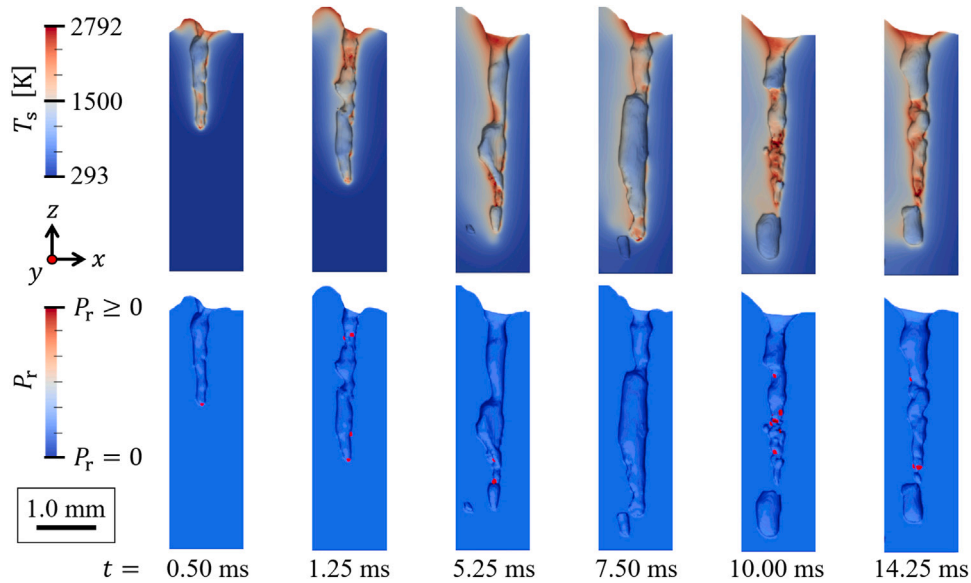


Fig. 12. Predicted temperature fields and the regions where a recoil pressure is applied to the keyhole wall for six different time steps during the laser welding process for parameter setup A3 ($P = 5000$ W, $v_r = 200$ mm/s, core = 100%). The upper figures show the predicted temperature fields as contour plots, while the lower figures highlight in red the regions where the applied recoil pressure is larger than zero. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

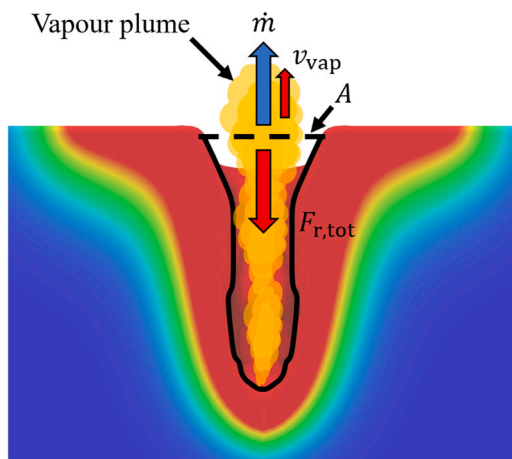


Fig. 13. Simplified model for the total recoil force exerted on the specimen, considering the net mass flow due to vaporization.

due to the escaping atoms during vaporization. Moreover, this recoil pressure is only applied to surfaces where vaporization occur. Actually, the formed vapour also contributes to the total recoil force, as it rises and exits the keyhole. Therefore, the total recoil force, $F_{r,tot}$, exerted on the specimen was calculated based on the predicted net mass flow due to vaporization, according to

$$F_{r,tot} = 2\dot{m}v_{vap} \quad (18)$$

where $\dot{m}$ is the predicted net mass flow due to vaporization and v_{vap} is the velocity of the vapour. A factor of two is included in Eq. (18) due to modelling only half of the weld seam. The vapour velocity is expressed in terms of the net mass flow, according to Eq. (19), where ρ_{vap} is the vapour density estimated from the ideal gas law according to Eq. (20), and A is the cross-sectional area of the keyhole opening, as shown in Fig. 13.

$$v_{vap} = \frac{\dot{m}}{\rho_{vap}A} \quad (19)$$

$$\rho_{vap} = \frac{P_{vap}M}{RT_{vap}} \quad (20)$$

It is worth noting that A represents half of the keyhole opening area, as the symmetry of the weld seam is utilized, and only half of the weld seam is modelled. This semicircular area was calculated with an average keyhole radius of 0.1 mm for the parameter setups with 100% peak power delivered to the core (A1-A3, A5, B1-B3, and B5) and 0.125 mm for the parameter setups with 65% peak power delivered to the core and an additional ring-shaped beam (A4 and B4). In Eq. (20), P_{vap} corresponds to the vapour partial pressure, M is the molar mass of the vaporized material, $R = 8.314 \text{ J/mol}$ is the universal gas constant, and T_{vap} is the vapour temperature. In this model, the molar masses for Al and Cu were considered as 26.98 g/mol and 63.55 g/mol, respectively. By combining Eq. (18) and Eq. (19) the total recoil force exerted on the specimen can be expressed as

$$F_{\text{r,tot}} = 2 \frac{\dot{m}^2}{\rho_{\text{vap}} A} \quad (21)$$

The net mass flow, $\dot{m}$, was calculated for each time step for the complete computational domain, based on the mass flow per unit area, $\dot{m}_a$, given by Eq. (13). The second term on the right-hand side of Eq. (13) contains the vapour partial pressure, P_{vap} , which controls the condensation rate in the model. However, by assuming that P_{vap} is not increased for higher laser powers leads to an increased force which is not seen in the experiments. To elaborate on that, P_{vap} was used as a model parameter to investigate how it affects the net mass flow and the recoil force. Here, the vapour partial pressure was assumed to increase with increasing laser power for the parameter setups with 100% peak power delivered to the core (A1-A3, A5, B1-B3, and B5), as shown in Eq. (22), where the vapour partial pressures are given in kPa.

$$\begin{aligned} P_{\text{vap}}^{(\text{Al})} &= 50P - 50, & 3 \text{ kW} \leq P \leq 5 \text{ kW} \\ P_{\text{vap}}^{(\text{Cu})} &= 15P + 50, & 4 \text{ kW} \leq P \leq 6 \text{ kW} \end{aligned} \quad (22)$$

This is based on the assumption that the condensation rate increases with increasing laser power. As a result of this assumed behavior, the net mass flow is limited following Eq. (13). For the parameter setups with 65% peak power delivered to the core (A4 and B4), a lower vapour partial pressure was assumed due to the larger keyhole opening resulting from the increased spot size from the additional ring-shaped beam. This assumption was made because the increased keyhole opening allows the vapour to escape from the keyhole more easily, resulting in a lower vapour partial pressure. The vapour partial pressures for parameter setups A4 and B4 were assumed as 100 kPa and 104 kPa, respectively.

Fig. 14a–b show the cumulative mass loss due to vaporization from the computational domain for parameter setups A1-A5 and B1-B5, respectively. From the cumulative mass loss in Fig. 14a–b, $\dot{m}$ was calculated for each parameter setup by determining the mean slope of the corresponding curve. As seen in Fig. 14a–b, $\dot{m}$ (or the net mass flow) increases with increasing laser power when the feed rate is held constant at 200 mm/s (parameter setups A1-A3 and B1-B3). When the laser power is held constant with an increased feed rate to 400 mm/s (parameter setups A5 and B5), the net mass flow decreases due to a corresponding reduction in energy input. For parameter setups A4 and B4, corresponding to the beam shapes with an additional ring-shaped beam, a significantly higher net mass flow is predicted, compared to the net mass flow for the same laser power and feed rate but without the ring-shaped beam.

After determination of the net mass flow ($\dot{m}$), the total recoil forces exerted on the specimen were calculated based on Eq. (21). Fig. 15a–b present the predicted recoil forces for parameter setups A1-A5 and B1-B5, respectively. Furthermore, a comparison between predicted and experimentally measured recoil forces is given here. For the experimentally measured recoil forces, the mean values for three repeated experiments for each parameter setup are shown, and the corresponding minimum and maximum values are illustrated as error bars. The experimental measurements show a consistent recoil force for increasing laser power and constant feed rate for the parameter setups with

100% peak power delivered to the core, for the Cu specimens, corresponding to parameter setups B1-B3. For the Al specimens the recoil force is consistent for increasing laser power for parameter setups A1 and A3, but for A2 a lower recoil force was observed. Even though the recoil force is not the same for different laser powers for the Al specimens, a consistent force was predicted by the model, similar to the behaviour of the Cu specimens. For parameter setups A4 and B4, corresponding to the beam shapes with an additional ring-shaped beam, larger recoil forces were predicted as shown in Fig. 15a–b. For parameter setups A5 and B5 experimentally measured recoil forces are missing, but the corresponding predicted values are shown in Fig. 15a–b.

4. Discussion

The relative errors for the Cu model is larger compared with the Al model. This is presumed to result from larger uncertainties in the laser-material interaction and the behaviour of the gaseous phase for the Cu material. Furthermore, these large relative errors are due trade-offs made in d_p and w_s during the model calibration for the relatively wide parameter range in v_f .

The experimental results presented in Section 3.1 shows that the process-induced recoil force is significantly increased when 65% of the peak power is delivered to the core fibre and an additional ring-shaped beam is used, compared to only using the core fibre. Both Al and Cu specimens exhibit this behaviour. If the total recoil force acting on the specimen is considered to result from the vapour escaping through the keyhole opening, then this increased recoil force may be explained by the larger keyhole opening associated with the use of a superimposed beam shape, consisting of both core and ring beams. This enlarged keyhole opening will allow the vapour to escape more easily and result in a lower vapour partial pressure inside the keyhole. This decreased vapour partial pressure will then result in a decreased condensation rate and thus a larger recoil force.

Furthermore, the experimental investigation shows that for a constant feed rate of 200 mm/s, the total recoil force is fairly consistent for an increasing laser power when 100% of the peak power is delivered to the core fibre, both for Al and Cu specimens. Since the vaporization rate should increase with an increased energy input to the base material, this behaviour may be explained by that the condensation rate also increases with increasing energy input and thus compensates for the increased rate of vaporization.

The experimentally measured recoil forces for the considered parameter setups for Al and Cu specimens ranges between 0.59 mN - 1.72 mN, and 0.52 mN - 1.22 mN, respectively, which have the same order of magnitude as the measured forces in for example [25,26]. However, the measured recoil forces in the present study are one order of magnitude lower than those measured in [27] for keyhole laser welding of stainless steel. Furthermore, the measurements in [27] demonstrates an increasing recoil force with increasing laser power, which have not been observed in the present study. This may be explained by the differences in thermo-physical properties and motivates further studies of different materials.

In the numerical investigation presented in Section 3.3, the proposed CFD models were used to investigate the relationships between recoil forces and various process parameters. Since the gaseous phases of the base material were not considered in the proposed CFD models, and the effects of vaporization were only modelled as boundary effects on the melt pool surface and keyhole wall, the total recoil force on the specimen was underestimated by only considering the recoil pressure acting at regions where vaporization occur. As this is a common simplification when modelling laser welding processes in order to reduce model complexity, it is essential to understand the effects of these assumptions. For example, the vapour partial pressure is not considered in most models but will probably affect the keyhole size and shape. As a result of these simplifications, the absence of this vapour partial

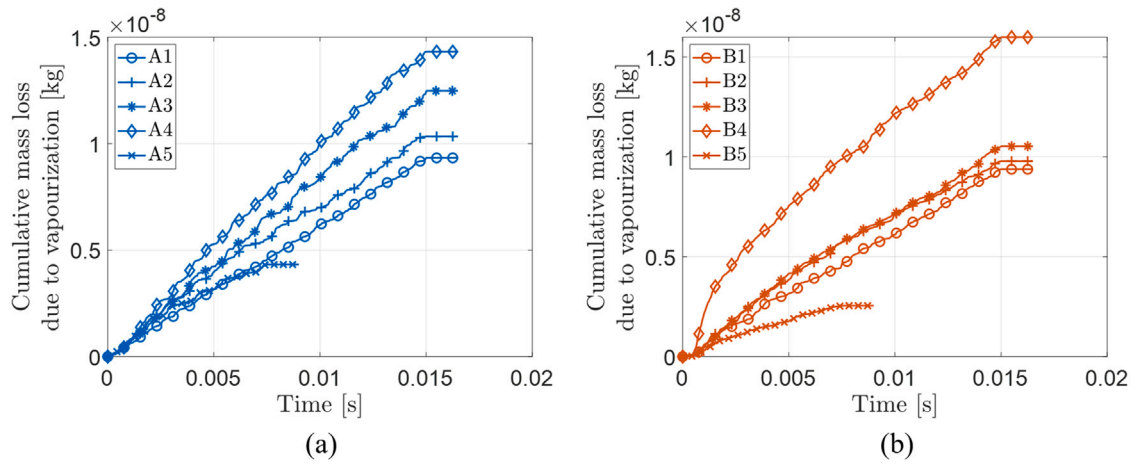


Fig. 14. Predicted cumulative mass loss due to vaporization from the computational domain (given in kg) for: (a) parameter setups A1-A5, and (b) parameter setups B1-B5.

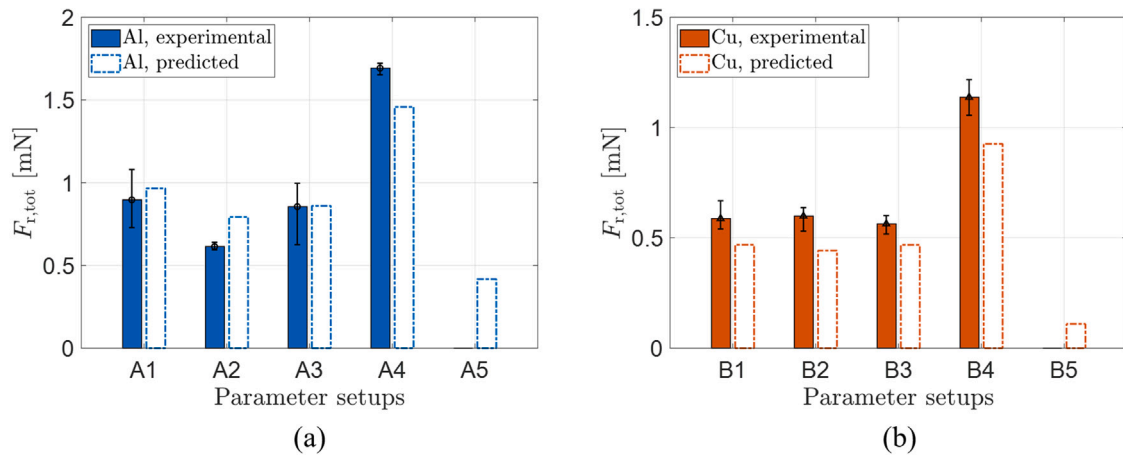


Fig. 15. Comparison of experimentally measured and predicted total recoil forces exerted on the specimen for: (a) parameter setups A1-A5, and (b) parameter setups B1-B5. For the experimentally measured recoil forces, the mean values for three repeated experiments for each parameter setup are shown, and the corresponding minimum and maximum values are illustrated as error bars.

pressure needs to be compensated by adjusting parameters outside their actual physical value in order to reflect experimental observations. It is worth to notice, however, that the large discrepancy in experimentally measured and numerically predicted recoil force might be due to other reasons as well, such as spatter or process instabilities. For example, the spread in measured recoil force, illustrated in Fig. 15, may be due to process instabilities such as spatter. For the A4 parameter setup it is clearly seen that the spread in the measured force is very low which may be explained by the use of the ring-core beam shape which usually produce stable welding conditions.

To account for that the gas formed due to vaporization also contribute to the total recoil force as it rises and exits the keyhole, the total recoil force was calculated based on the predicted net mass flow due to vaporization. In this approach, the CFD models were first calibrated and validated by comparing the predicted joint geometries with those extracted from experimentally obtained micrographs. The models were then used in an attempt to explain the experimentally measured recoil forces for the considered parameter setups. As the experimental measurements for the specimens produced with the core dominated beam show a fairly consistent recoil force for increasing laser power and constant feed rate, the condensation rate was assumed to increase with increasing laser power in order to compensate for the corresponding increase in vaporization rate. In the CFD models, this was achieved by assuming that the vapour partial pressure is linearly increased with increasing laser power. This increased vapour partial pressure will result in a decreased net mass flow, according to Eq. (13),

which can be seen as an increased condensation rate. Moreover, this increased vapour partial pressure results in an increased vapour density, and this increased density together with the decreased net mass flow results in lower recoil force, according to Eq. (21). For the specimens produced with the superimposed beam, consisting of both core and ring beams, a lower vapour partial pressure was assumed. This is due to the larger keyhole opening associated to the superimposed beam, which allows the vapour to escape more easily from the keyhole. The lower recoil forces associated with larger feed rates may be explained by the decreased net mass flow associated with the reduction in energy input resulting from an increased feed rate.

As seen in Fig. 14a–b, the net mass flow increases with increasing laser power when the feed rate is held constant. The increase in net mass flow is attributed to the increase in energy input to the base material due to the increased laser power. However, the assumed increase in vapour partial pressure with increasing laser power results in a smaller increase of the net mass flow compared to assuming a constant vapour partial pressure. This is because a higher vapour partial pressure reduces the net mass flow, as shown in Eq. (13), corresponding to a higher condensation rate. For the beam shapes with an additional ring-shaped beam, a significantly higher net mass flow was predicted. This is due to the assumed lower vapour partial pressure for these parameter setups.

As seen in Fig. 15a–b, this simplified model, together with the assumptions made for the vapour partial pressure, successfully captures the general behaviour of the experimentally measured recoil forces

for the considered parameter setups. However, for the Cu model the predicted values follows the same trend as the experimental values but are consistently lower for all parameter setups. This may be due to the assumptions made for the keyhole opening area, the vapour temperature inside the keyhole or the vapour partial pressure. It is worth noting, however, that in this simplified approach, where the gaseous phases are not directly considered, a constant vapour partial pressure was assumed for each individual parameter setup, even though this pressure was varied between the different parameter setups. This differs from the approach used by for example [15], where the vapour partial pressure, and thus the condensation rate, is an implicit result of the modelled gaseous phases. However, the approach in the present study allows for a less complex and more time-efficient model.

Before any conclusions are made from these experimental and numerical investigations, it is important to acknowledge some assumptions and limitations that may have influenced the outcomes and findings:

- The assumed vapour temperatures for the Al and Cu vapours of 6400 K and 8000 K, respectively, are based on the calibration process, which compares predicted and experimentally determined joint geometries. Although these temperatures are of the same order of magnitude as other observed vapour temperatures (cf. [1]), it should be noted that assuming these temperatures differently would affect the predicted total recoil force.
- The keyhole opening area, which was used to calculate the total recoil force, was calculated based on the average radius of the keyhole opening determined from the CFD simulations. Even though this average radius was calculated over multiple time steps, the choice of time steps may influence the magnitude of this average radius.
- The metal vapour formed during the laser welding process is assumed to behave as an ideal gas.
- In the analysis, it is assumed that the entire keyhole is filled with metal vapour, i.e. the partial pressure for the ambient air is assumed as zero inside the keyhole.
- The vapour partial pressure is assumed as homogeneously distributed inside the keyhole, which is a simplification from the real distribution that varies spatially.
- The vapour partial pressure is assumed to linearly increase with increasing laser power, but for the individual parameter setups a constant vapour partial pressure was assumed.
- In the CFD models, the net condensation is considered in regions only where vaporization occurs, which means that the assumed vapour partial pressures could be slightly overestimated in order to account for condensation in other regions.
- During the experiments the amount of spatter was not measured, but the process parameter ranges were selected to obtain fairly stable welding conditions.

5. Conclusions

In the present study, the process-induced recoil force during laser welding of aluminium (Al) and copper (Cu) specimens was experimentally measured. Furthermore, multi-physics computational fluid dynamics (CFD) models were developed for the laser welding process and calibrated using experimentally measured joint geometries. These models were used to predict and investigate the process-induced recoil force and its dependence on various process parameters. The following conclusions can be summarized for this investigation:

- (1) For the copper (Cu) specimens, the experimental measurements showed a consistent recoil force for increasing laser power and constant feed rate of 200 mm/s, when using a beam shape corresponding to 100% of the peak power delivered to the core fibre. For the copper (Cu) specimens, the measured recoil forces range from 0.52 mN to 0.67 mN, for a laser power range of 4000 W to 6000 W.

- (2) For the aluminium (Al) specimens, the experimental measurements showed a consistent recoil force for laser powers of 3000 W and 5000 W and constant feed rate of 200 mm/s, when using a beam shape corresponding to 100% of the peak power delivered to the core fibre. For a laser power of 4000 W a 32% lower recoil force was observed. For the aluminium (Al) specimens, the measured recoil forces range from 0.59 mN to 1.08 mN, for a laser power range of 3000 W to 5000 W.
- (3) With the use of an additional ring-shaped beam, with 35% of the peak power delivered to the ring fibre and 65% to the core fibre, the experimentally measured recoil forces were significantly increased, both for aluminium (Al) and copper (Cu) specimens. For this beam shape, the experimentally measured recoil forces range from 1.65 mN to 1.72 mN for the aluminium (Al) specimens, and 1.06 mN to 1.22 mN for the copper (Cu) specimens.
- (4) From the numerical investigation, the consistency in recoil force for increasing laser power may be explained by an increased vapour partial pressure, and thus an increased condensation rate, for increasing laser power.
- (5) The numerical investigation also shows that the larger recoil forces associated with a beam shape with an additional ring-shaped beam, may be explained by the corresponding larger keyhole opening which leads to a decreased vapour partial pressure and condensation rate, and consequently an increased mass flow out of the keyhole resulting in an increased recoil force.

These findings emphasize the critical role of understanding the behaviour of the vaporized gas phase in the keyhole in determining appropriate simplifications and assumptions for laser welding process modelling. Additionally, these findings contribute to the advancement of multi-physics process models, supporting their development and validation while enhancing knowledge of relevant modelling approximations. Although this study specifically examines laser welding of aluminium (Al) and copper (Cu), the insights gained can be extended to other materials, and the methodologies employed are applicable to a wide range of laser welding processes as well as laser-based additive manufacturing applications.

As future research work, a detailed numerical investigation is suggested, where multi-physics computational fluid dynamics (CFD) simulations considering the gaseous phases are used to investigate the influence of various process parameters on the vapour partial pressure and the condensation rate during the laser welding process. Moreover, the experimentally measured recoil forces from the present study, can be used in future research works as a reference for comparison and validation of predicted recoil forces from multi-physics simulation models. Furthermore, a more comprehensive analysis of the impact of various materials and beam shapes on the recoil force during keyhole laser welding are suggested as future work.

CRedit authorship contribution statement

Andreas Andersson Lassila: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Erik Lundell:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Tobias Andersson:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Dan Lönn:** Writing – review & editing, Visualization, Methodology, Investigation. **Kent Salomonsson:** Writing – review & editing, Supervision, Project administration. **Rohollah Ghasemi:** Writing – review & editing, Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used and/or analysed during the current study are available from the corresponding author on reasonable request.

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